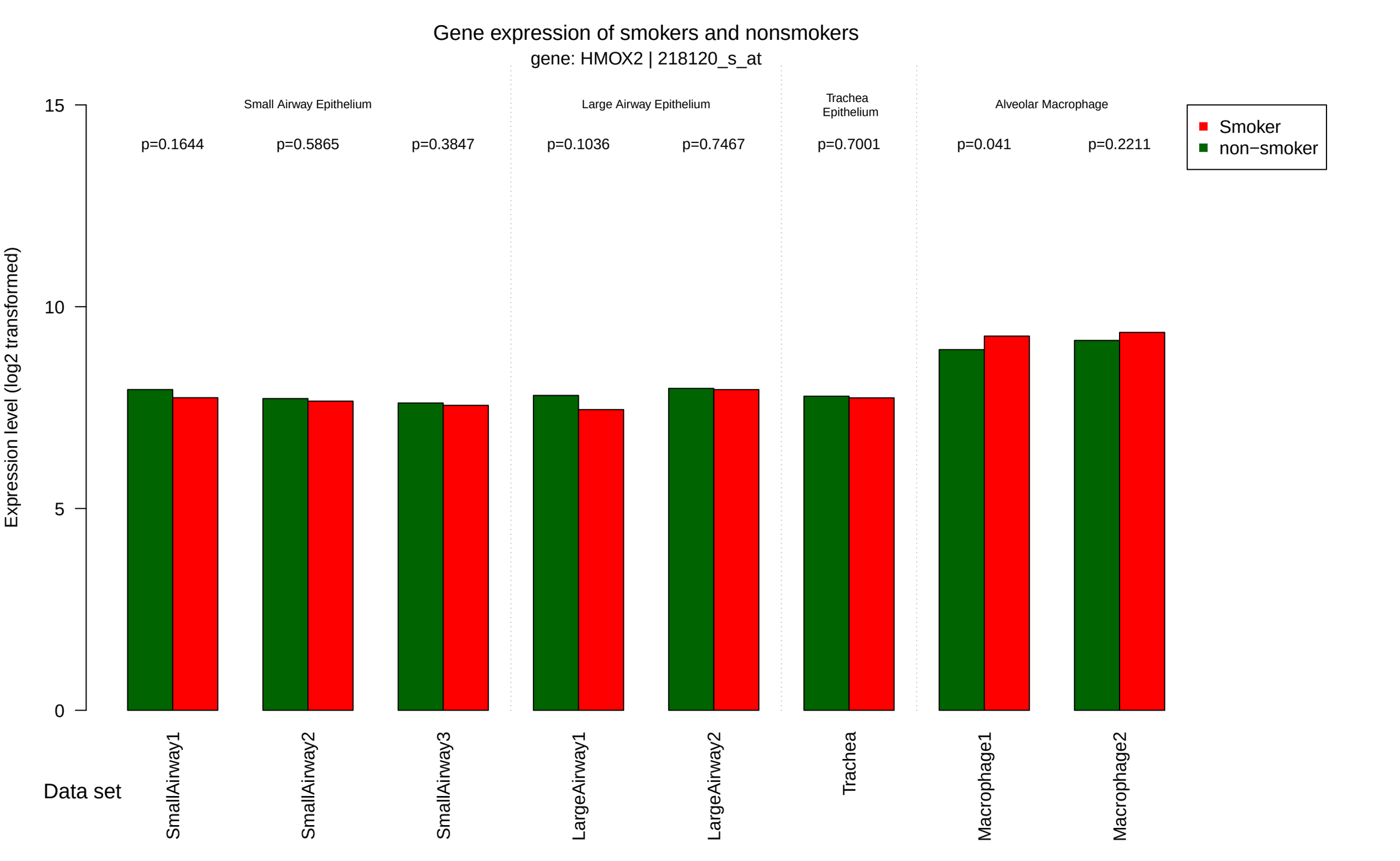
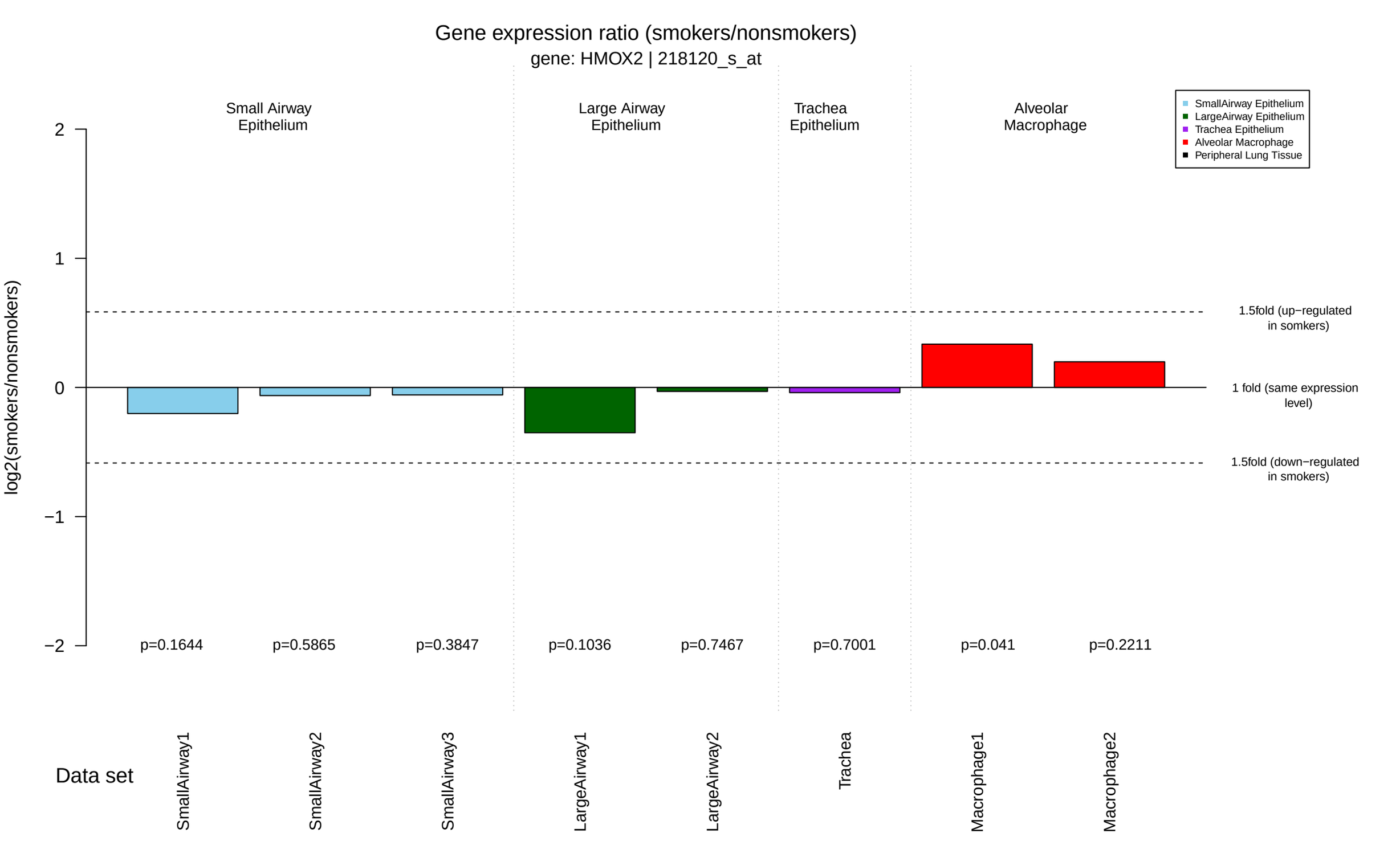
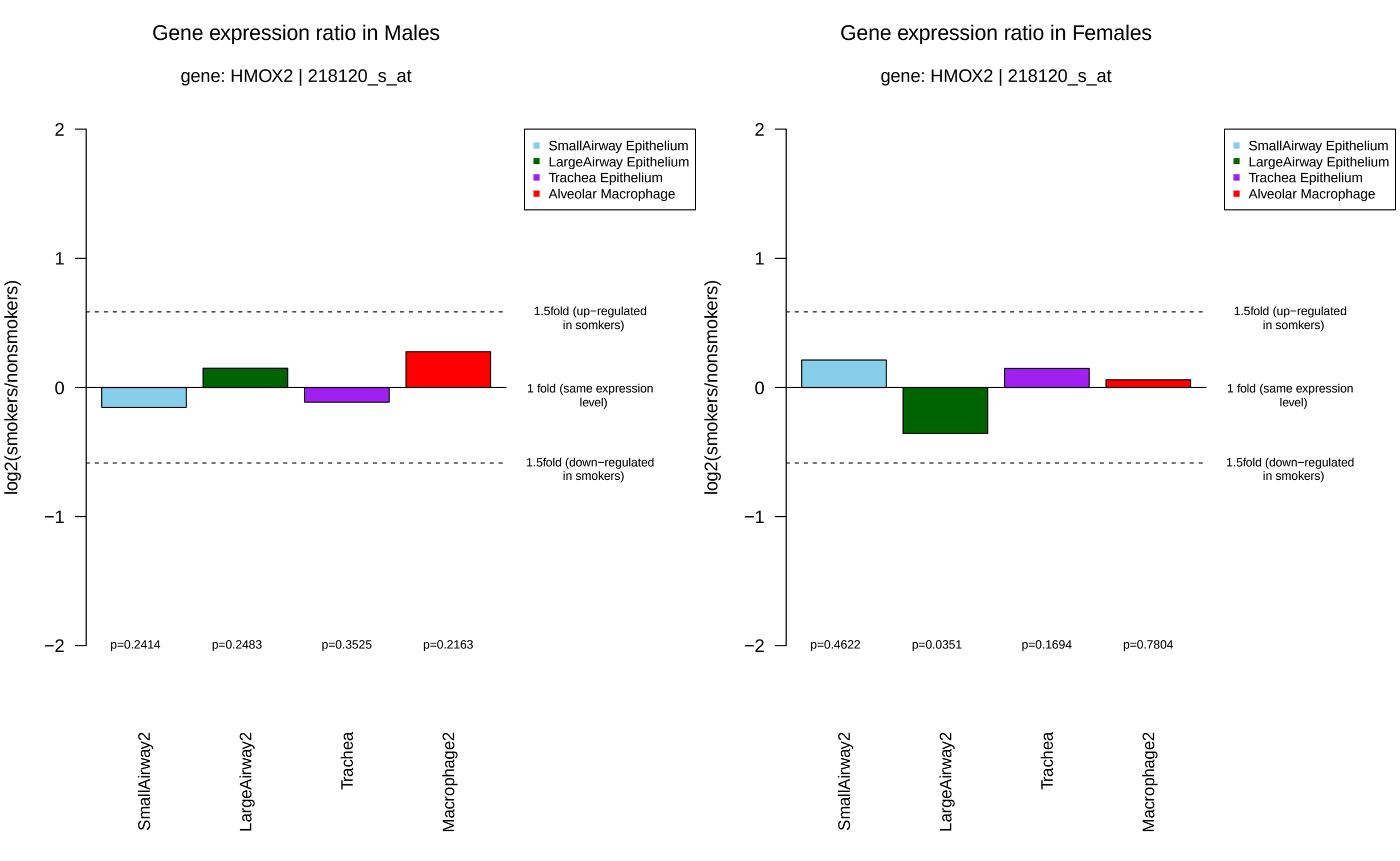
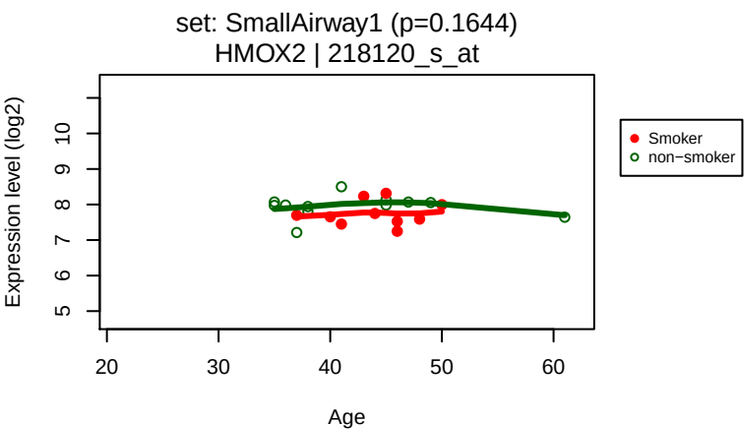
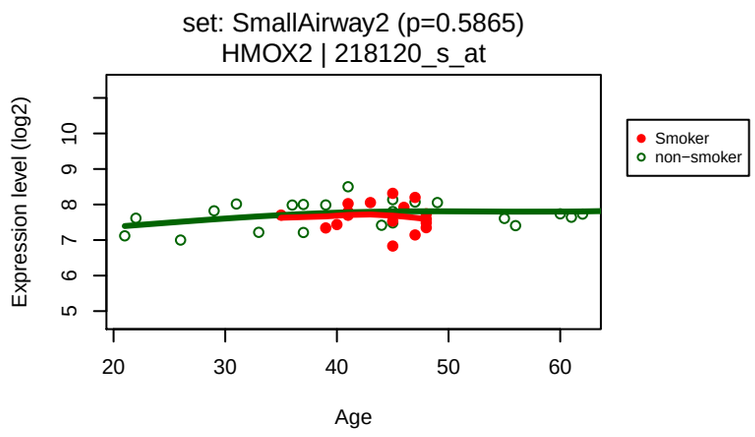
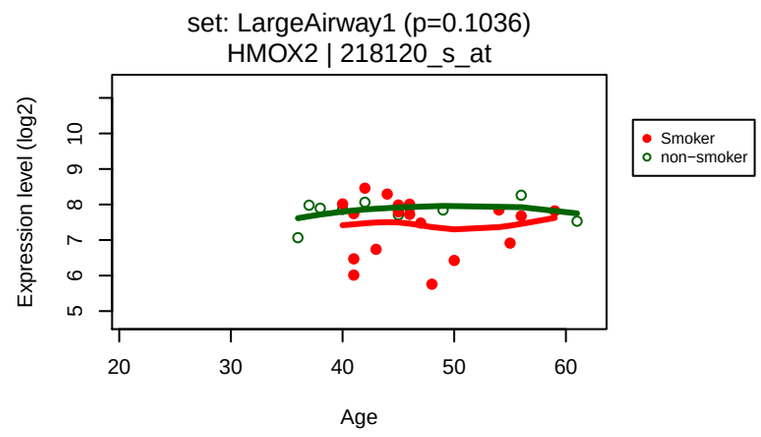

Gene expression of smokers and nonsmokers
gene: HMOX2 | 218120_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1644
p=0.5865
p=0.3847
p=0.1036
p=0.7467
p=0.7001
p=0.041
p=0.2211
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

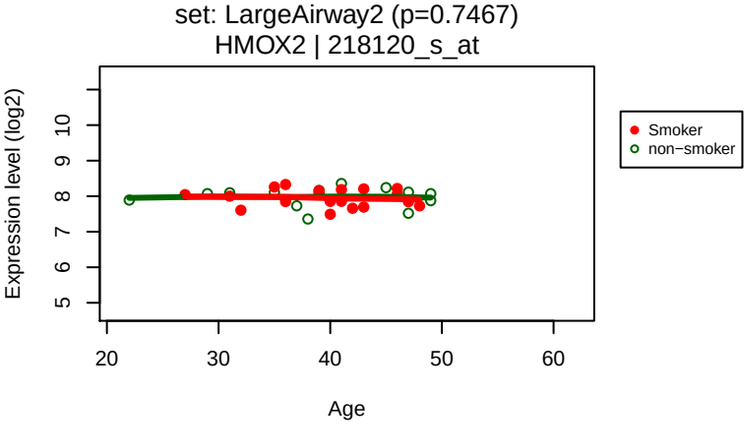

Gene expression ratio (smokers/nonsmokers)
gene: HMOX2 | 218120_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1644
p=0.5865
p=0.3847
p=0.1036
p=0.7467
p=0.7001
p=0.041
p=0.2211
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

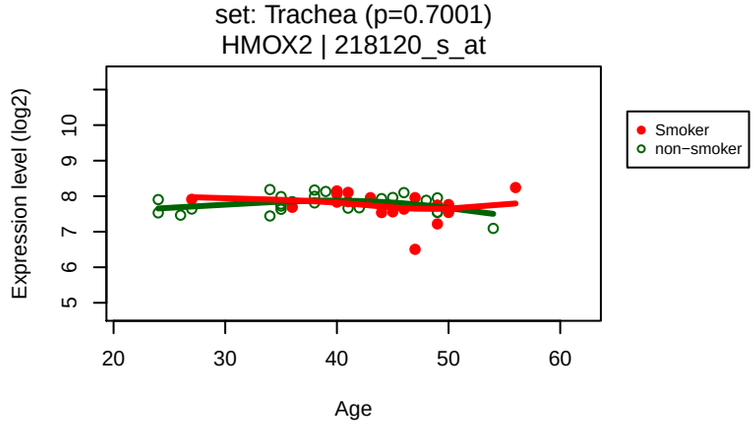

Gene expression ratio in Males
gene: HMOX2 | 218120_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2414
p=0.2483
p=0.3525
p=0.2163
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HMOX2 | 218120_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4622
p=0.0351
p=0.1694
p=0.7804
SmallAirway2
LargeAirway2
Trachea
Macrophage2

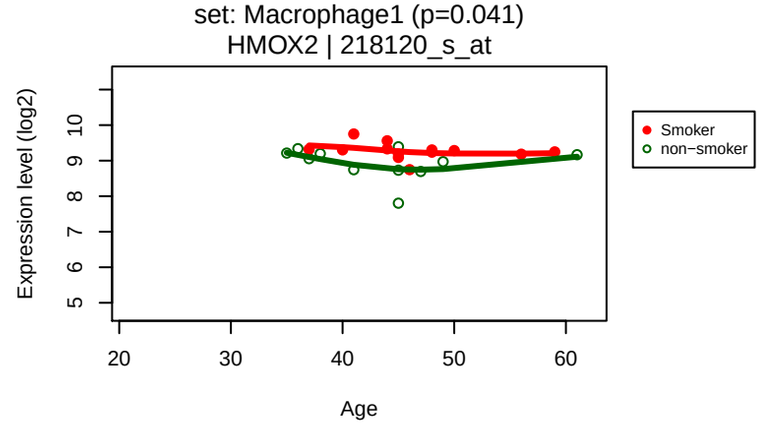

set: SmallAirway1 (p=0.1644)
HMOX2 | 218120_s_at
Expression level (log2)
Age
Smoker
non−smoker

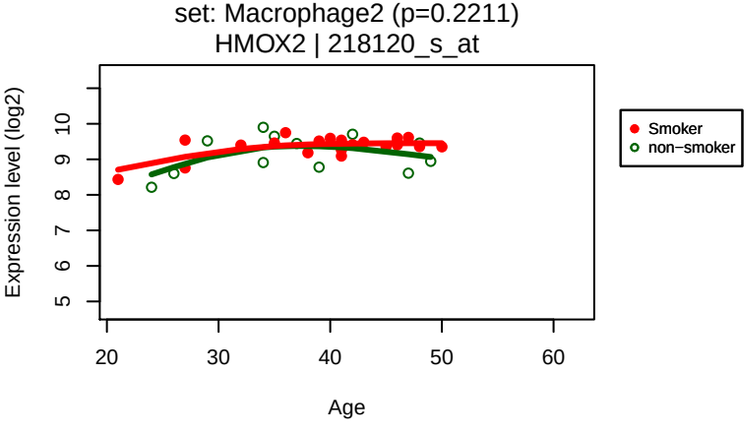

set: SmallAirway2 (p=0.5865)
HMOX2 | 218120_s_at
Expression level (log2)
Age
Smoker
non−smoker

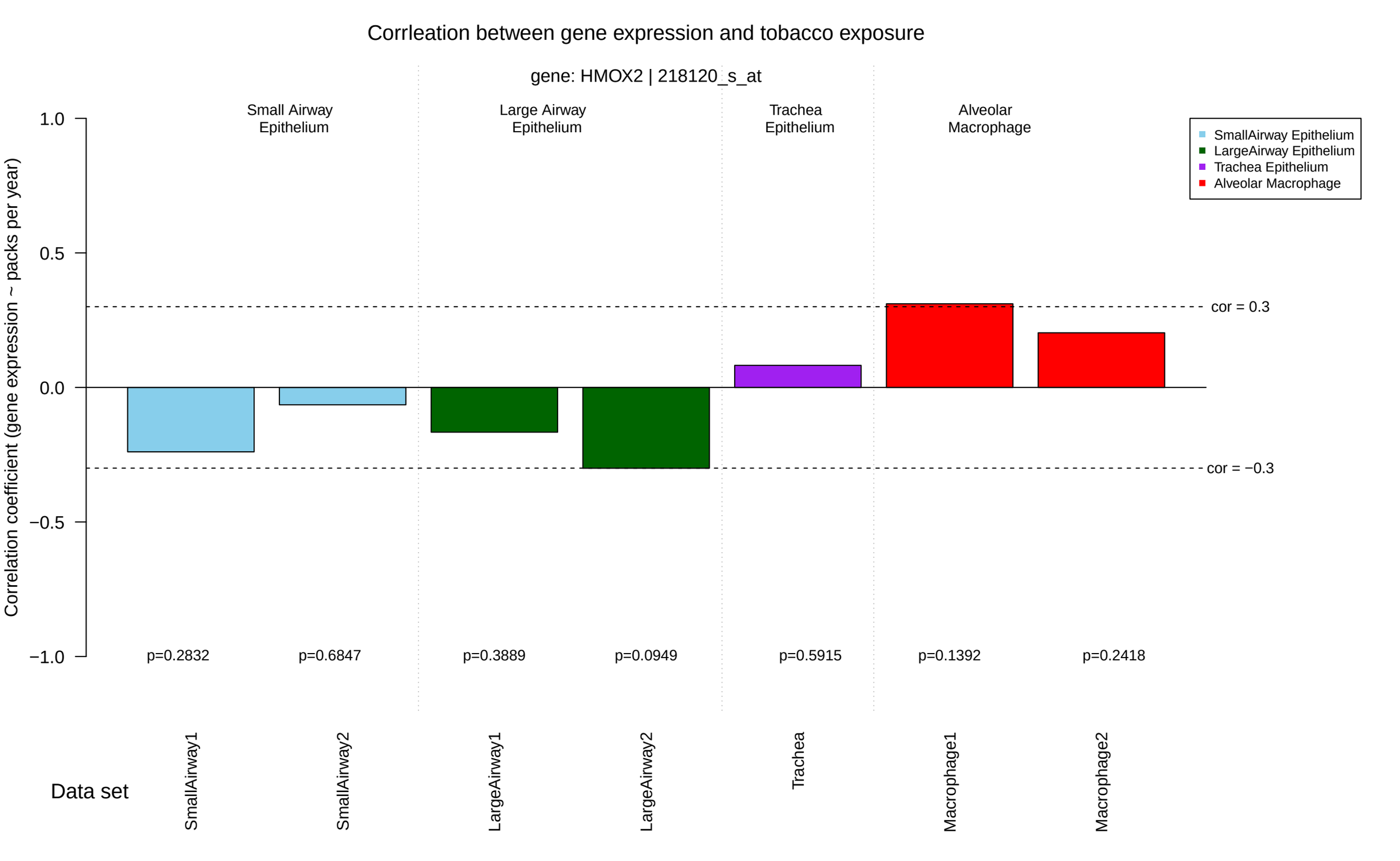

set: LargeAirway1 (p=0.1036)
HMOX2 | 218120_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7467)
HMOX2 | 218120_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7001)
HMOX2 | 218120_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.041)
HMOX2 | 218120_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2211)
HMOX2 | 218120_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HMOX2 | 218120_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2832
p=0.6847
p=0.3889
p=0.0949
p=0.5915
p=0.1392
p=0.2418
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2